IntSeqBERT: Dual-Stream Masked Modelling for Integer Sequences via Magnitude and Modular Arithmetic Embeddings

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Abstract. Integer sequences in the OEIS span values from single-digit constants to astronomical factorials and exponentials, making prediction challenging for standard tokenised models that cannot handle out-of-vocabulary values or exploit periodic arithmetic structure. We present **IntSeqBERT**, a dual-stream Transformer encoder for masked integer-sequence modelling on OEIS. Each sequence element is encoded along two complementary axes: a continuous log-scale Magnitude embedding and Sin/Cos Modulo embeddings for 100 residues (moduli 2–101), fused via FiLM. Three prediction heads (magnitude regression, sign classification, and Modulo prediction for 100 moduli) are trained jointly on 274,705 OEIS sequences. At the Large scale (91.5M parameters), IntSeqBERT achieves 95.85% Magnitude accuracy and 50.38% Mean Modulo Accuracy (MMA) on the test set, outperforming a standard tokenised Transformer baseline by +8.9 pt and +4.5 pt, respectively. An ablation removing the Modulo stream confirms it accounts for +15.2 pt of the MMA gain and contributes an additional +6.2 pt to Magnitude accuracy. A CRT-based IntegerSolver converts the model's predictions into concrete integers, yielding a 7.4× improvement in next-term prediction (Top-1: 19.09% vs. 2.59%). Modulo spectrum analysis reveals a strong negative correlation between Normalised Information Gain (NIG) and Euler's totient ratio $\varphi(m)/m$ ($r = -0.851$, $p < 10^{-28}$), providing empirical evidence that composite moduli capture OEIS arithmetic structure more efficiently via CRT aggregation.

Keywords: Integer sequences · OEIS · Masked sequence modelling · Modular arithmetic · Transformer · FiLM

1 Introduction

The On-Line Encyclopedia of Integer Sequences (OEIS) [14], as of January 2026, catalogues more than 391,000 entries spanning combinatorics, number theory, algebra, and many other branches of mathematics, making it the *de facto* standard reference for integer sequences. Each entry associates a finite integer sequence with its mathematical definition, rendering the OEIS a uniquely machine-readable corpus of mathematical knowledge.

The task we formalise is *masked sequence modelling*: a random subset of positions in a sequence is masked and the model is trained to predict each masked value from its surrounding context. This task directly probes how well a model has internalised the arithmetic and combinatorial laws governing integer sequences; next-term prediction is treated as a special case and evaluated via a dedicated Solver component. The prior benchmark FACT [1] defines and evaluates six tasks including unmasking, establishing what a plain Transformer can achieve, but it faces fundamental limitations in handling large integer values unseen at training time and in learning multiplicative structure.

The principal difficulty of this task is the extreme heterogeneity of integer sequences. Values range from single-digit constants to astronomically large factorials and exponential sequences, with differences of tens of orders of magnitude within a single training corpus. At the same time, many sequences obey *residue constraints*—parity, combinatorial patterns, or periodic remainders—that are independent of the magnitude of individual values. The standard tokenisation approach of assigning each integer a discrete vocabulary token is fundamentally ill-suited to this setting: it cannot handle large integers unseen at training time, embeds arithmetic structure in opaque token IDs, and breaks down in scale for large numbers.

We propose **IntSeqBERT**, a Transformer encoder pre-trained on the OEIS corpus via masked sequence modelling, which overcomes the above challenges through a *dual-stream* input representation. Rather than tokenising integers, IntSeqBERT encodes each element along two complementary axes:

- **Magnitude stream**: a continuous logscale embedding of the absolute value, capturing growth behaviour and scale.
- **Modulo stream**: Sin/Cos embeddings for 100 residues modulo 2 through 101, capturing periodicity and number-theoretic structure.

The two streams are fused via FiLM (Feature-wise Linear Modulation) [10]. During training, three predictors are jointly optimised in a multi-task objective: magnitude regression, sign classification, and Modulo prediction for 100 moduli. Concrete integer values are recovered from the masked-position predictions (magnitude, sign, residue distributions) using a Chinese Remainder Theorem (CRT) based **IntegerSolver** (Section 3.9).

We evaluate IntSeqBERT against two baselines on a dataset of 274,705 OEIS sequences across three model sizes (Small: ∼9M, Middle: ∼44M, Large: ∼110M parameters). All experiments are conducted on a single GeForce RTX 3070 Ti (8 GB VRAM), and results should be interpreted in light of this consumer-GPU memory constraint. Our main findings are:

- Large-scale IntSeqBERT achieves **95.85%** Magnitude accuracy and **50.38%** Mean Modulo Accuracy (MMA) on the test set, surpassing the standard tokenised Transformer (Vanilla) baseline by +8.9 pt and +4.5 pt, respectively.
- The dual-stream representation yields a **7.4**× improvement in exact-match accuracy on next-term prediction via the Solver (Top-1: 19.09% vs. 2.59%).

– Modulo spectrum analysis reveals that NIG (Normalised Information Gain) is strongly negatively correlated with Euler's totient ratio $\varphi(m)/m$ ($r = -0.851$, $p < 10^{-28}$), providing empirical evidence that composite moduli aggregate arithmetic structure of OEIS sequences more efficiently via the CRT.
– Modulo accuracy and Solver accuracy improve more steeply with model scale than Magnitude accuracy, suggesting that arithmetic reasoning benefits disproportionately from increased model capacity.

Contributions.

1. *IntSeqBERT*: a dual-stream Transformer architecture for integer sequences that fuses magnitude and modular-arithmetic feature embeddings via FiLM.
2. *Multi-task pre-training on OEIS*: combining magnitude regression, sign classification, and 100-dimensional Modulo prediction in a single objective.
3. *Comprehensive evaluation*: scale-stratified Magnitude analysis, Modulo spectrum analysis, and Solver-based next-term prediction.
4. *Number-theoretic insight*: empirical discovery of a strong negative correlation ($r = -0.851$) between NIG and $\varphi(m)/m$, quantitatively demonstrating that composite moduli capture periodic arithmetic structure of OEIS sequences more efficiently via CRT aggregation.

Paper organisation. Section 2 reviews related work. Section 3 presents the IntSeqBERT architecture. Section 4 describes the dataset and experimental setup. Section 5 reports experimental results. Section 6 provides ablation and scaling analyses. Section 7 concludes.

2 Background and Related Work

2.1 AI for Mathematics

Throughout the history of mathematics, observing numerical similarities among sequence coefficients has seeded important conjectures. Landmark examples include the Monstrous Moonshine conjecture [4], the BSD conjecture motivated by numerical experiments on ranks of elliptic curves [2], the proof of Fermat's Last Theorem via the Shimura–Taniyama conjecture [18], and Candelas et al.'s prediction of 317,206,375 rational curves of degree 3 from mirror symmetry [3]. Numerical pattern observation can thus serve as a source of mathematical discovery; automating this process at scale with AI could contribute to finding new conjectural candidates.

Recent years have seen rapid progress in AI for mathematics. On problem-solving and proof, AlphaGeometry [15] automatically solves IMO geometry problems at high accuracy, and AlphaProof [9] resolved four of six problems at IMO 2024 in Lean formal proofs. On formalisation, Urban's autoformalization research [16] is advancing towards automatic translation of mathematical text into machine-verifiable form. On conjecture generation from numerical patterns, the Ramanujan Machine [11] automatically conjectures continued-fraction identities for fundamental constants such as π and e, with some subsequently proved by mathematicians. FunSearch [13] combines large language models with evolutionary search to discover constructions exceeding known bounds for the cap-set problem.

While these achievements demonstrate the viability of AI-driven mathematical discovery, the process of automatically extracting laws from numerical sequence patterns across diverse mathematical domains remains largely unexplored. This work aims to establish the representational foundations for such a capability.

2.2 AI Research on OEIS

Urban et al.'s Alien Coding [6] proposes a program-synthesis framework that, given an OEIS sequence, searches for the shortest code reproducing it. Follow-up work QSynt [7] by Gauthier and Urban introduces efficient program synthesis via self-learning tree search, successfully generating programs for 43,516 OEIS sequences. Learning Conjecturing from Scratch [8] further proposes a self-learning feedback method for inductive conjecture generation, automatically proving 5,565 arithmetic problems.

FACT [1] (Belčák et al., NeurIPS 2022) constructs a comprehensive benchmark on OEIS, defining six tasks—classification, similarity, sequence-part prediction, next-element extrapolation, *unmasking*, and continuation—and evaluating MLP, RNN, CNN, and Transformer baselines. A key finding is that the unmasking task is the most challenging of the six. This work directly targets unmasking as its primary objective. Our Vanilla baseline corresponds to FACT's plain Transformer, and IntSeqBERT builds upon it with Modulo feature engineering and FiLM fusion to substantially improve unmasking accuracy.

While Alien Coding aims to explicitly describe the generation rule for each individual sequence via program synthesis, FACT and this work are grounded in representation learning. Where FACT established what a plain Transformer can achieve, this work asks how far feature engineering can push that frontier.

2.3 Deep Learning and Arithmetic: Spectral Bias

Deep networks exhibit a well-known *spectral bias* [12]: they learn low-frequency functions before high-frequency ones. In the context of integer sequences, this is particularly problematic for multiplicatively growing sequences (e.g., n^2, $n!$), where learning the growth law from token sequences alone requires many layers. This work mitigates the problem through feature engineering: because the results of multiplication appear naturally in modular residues, supplying the Modulo spectrum as explicit input features reduces the network depth required to "rediscover" multiplicative structure.

2.4 Why Modulo Spectra? Number-Theoretic Motivation

Most OEIS sequences are generated by finite algorithms combining addition, multiplication, and modular operations. Crucially, modular arithmetic is a *ring homomorphism*:

$$(a+b) \bmod m = \big((a \bmod m) + (b \bmod m)\big) \bmod m, \tag{1}$$

$$(a \times b) \bmod m = \big((a \bmod m) \times (b \bmod m)\big) \bmod m. \tag{2}$$

Hence, the residue sequence $(x_i \bmod m)$ faithfully preserves the additive and multiplicative structure of the original sequence as a compact representation, regardless of the magnitudes involved.

Fermat's little theorem implies that $(a^n \bmod p)$ is periodic with period dividing $p-1$ for prime p with $\gcd(a,p) = 1$. Gauss's law of quadratic reciprocity [5] and its generalisations further link residue information across different primes. Composite moduli simultaneously retain information from all prime power factors via the Chinese Remainder Theorem (CRT), motivating the use of both prime and composite moduli in our spectrum. Our Modulo features cover $m \in \{2, 3, \dots, 101\}$ (100 moduli), capturing power-residue structure broadly across primes and composites.

2.5 Neural Representations of Numbers

Approaches to neural number representation include (i) tokenisation, the LLM mainstream, which cannot handle out-of-vocabulary values and buries arithmetic structure in opaque token IDs; (ii) log-scale continuous embeddings, which improve scale invariance but cannot capture arithmetic periodicity; and (iii) sinusoidal embeddings, as used for positional encoding in Transformers [17], which exploit periodicity. In this work we repurpose sinusoidal embeddings from *positions* to *modular residues of values*. FiLM [10], originally proposed for conditioning visual features on language features in visual QA, is reused here to modulate the Magnitude stream via the Modulo stream, allowing arithmetic periodicity to condition continuous-value regression.

2.6 Positioning of This Work

In summary, the contribution of this work can be positioned as follows. While the Urban group's programme of Alien Coding [6], QSynt [7], and Learning Conjecturing [8] aims at *program synthesis* and conjecture generation for individual sequences, this work pursues *representation learning* to acquire, in a shared latent space, the arithmetic structure common to the entire OEIS corpus. Where FACT [1] identified the limits of a plain tokenised Transformer (out-of-vocabulary values, scale collapse, failure to learn multiplicative structure), this work overcomes those limits by adding Modulo-spectrum feature engineering and FiLM fusion. The resulting dual-stream design can also be viewed as an architectural implementation of the mathematical knowledge-acquisition process of moving from numerical-pattern observation to law extraction, as described in Section 2.1.

3 IntSeqBERT

3.1 Problem Formulation

Let $\mathbf{x} = (x_1, x_2, \ldots, x_L)$ with $x_i \in \mathbb{Z}$ and $L \le 128$ be a finite prefix of an OEIS sequence. We adopt *masked sequence modelling*: a random subset of positions is masked and the model is trained to predict each masked value. Specifically, for each masked position i, three quantities are predicted:

1. **Magnitude**: $v_i = \begin{cases} 0 & (x_i = 0), \\ 1 + \log_{10}|x_i| & (x_i \neq 0) \end{cases} \in \mathbb{R}_{\ge 0}$
2. **Sign**: $s_i \in \{+, -, 0\}$ (3-class label)
3. **Residues**: $r_i^{(m)} = x_i \bmod m$ for each $m \in \{2, 3, \ldots, 101\}$ (100 independent classification targets)

This decomposition separates magnitude, sign, and periodic arithmetic structure into complementary supervision signals.

3.2 Input Feature Extraction

Two feature vectors are computed for each element x_i prior to learnable embeddings.

Magnitude features $\mathbf{f}_i^{\text{mag}} \in \mathbb{R}^4$:

$$\mathbf{f}_i^{\text{mag}} = \big[v_i,\ \mathbf{1}[x_i > 0],\ \mathbf{1}[x_i < 0],\ \mathbf{1}[x_i = 0]\big].$$

The last three components are a one-hot sign representation. For astronomically large integers exceeding the `float64` range, $|x_i|$ is replaced by its decimal digit count.

Modulo features $\mathbf{f}_i^{\text{mod}} \in \mathbb{R}^{200}$: For each modulus $m \in \{2, \ldots, 101\}$, let $r = x_i \bmod m$, and embed the residue as a point on the unit circle:

$$\phi_m(r) = \left[\sin\!\left(\tfrac{2\pi r}{m}\right),\ \cos\!\left(\tfrac{2\pi r}{m}\right)\right] \in \mathbb{R}^2.$$

Concatenating all 100 moduli gives $\mathbf{f}_i^{\text{mod}} \in \mathbb{R}^{200}$. This Sin/Cos embedding is equivariant to the group structure of $\mathbb{Z}/m\mathbb{Z}$ and avoids discontinuities at wrap-around boundaries.

3.3 Dual-Stream Embedding

The two feature vectors are projected to the model hidden dimension d by independent projection layers. A two-layer MLP is used for the Magnitude side:

$$\mathbf{h}_i^{\text{mag}} = \text{MLP}_{\text{mag}}(\mathbf{f}_i^{\text{mag}}), \quad \mathbf{h}_i^{\text{mod}} = W_{\text{mod}}\,\mathbf{f}_i^{\text{mod}} + \mathbf{b}_{\text{mod}}, \quad \mathbf{h}_i^{\text{mag}}, \mathbf{h}_i^{\text{mod}} \in \mathbb{R}^d.$$

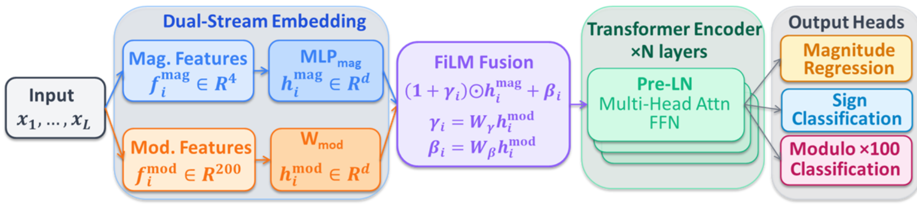

Fig. 1. IntSeqBERT architecture. The Dual-Stream Embedding block projects $\mathbf{f}_i^{\mathrm{mag}} \in \mathbb{R}^4$ and $\mathbf{f}_i^{\mathrm{mod}} \in \mathbb{R}^{200}$ to $\mathbb{R}^d$ via $\mathrm{MLP}_{\mathrm{mag}}$ and W_{mod}, then fuses them with FiLM: $\mathbf{e}_i = (1 + \boldsymbol{\gamma}_i) \odot \mathbf{h}_i^{\mathrm{mag}} + \boldsymbol{\beta}_i$. Positional encodings are added before the Pre-LN Transformer encoder. Three prediction heads produce $\hat{v}, \hat{\sigma}^2 \in \mathbb{R}$ (magnitude regression), $\hat{s} \in \{+, -, 0\}$ (sign classification), and $\hat{r}^{(m)} \in \{0, \ldots, m-1\}$ (100× Modulo classification).

Table 1. Model configurations.

Config	Layers	d	Heads	Parameters
Small	6	256	4	6.4M
Middle	8	512	8	29.0M
Large	12	768	12	91.5M

3.4 FiLM Fusion

The two streams are fused by Feature-wise Linear Modulation (FiLM) [10]. The Modulo embedding generates element-wise scale $\boldsymbol{\gamma}_i$ and shift $\boldsymbol{\beta}_i$ to modulate the Magnitude embedding:

$$\boldsymbol{\gamma}_i = W_\gamma \, \mathbf{h}_i^{\mathrm{mod}}, \quad \boldsymbol{\beta}_i = W_\beta \, \mathbf{h}_i^{\mathrm{mod}},$$

$$\mathbf{e}_i = (1 + \boldsymbol{\gamma}_i) \odot \mathbf{h}_i^{\mathrm{mag}} + \boldsymbol{\beta}_i.$$

A ReLU is applied after the Modulo projection, and dropout is inserted before FiLM. This formulation allows arithmetic periodicity to condition continuous-value Magnitude representations with parameter efficiency ($W_\gamma, W_\beta \in \mathbb{R}^{d \times d}$). Standard sinusoidal positional encodings are added to $\mathbf{e}_i$ before the encoder.

The overall architecture is illustrated in Fig. 1: each element is projected independently via the Magnitude stream (blue) and the Modulo stream (orange), fused by FiLM, and passed to the Transformer encoder.

3.5 Transformer Encoder

The fused sequence $(\mathbf{e}_1, \ldots, \mathbf{e}_L)$ is processed by a standard Transformer encoder [17] with Pre-Layer Normalisation [19]. Three model sizes are evaluated:

3.6 Prediction Heads

Let $\mathbf{z}_i \in \mathbb{R}^d$ be the encoder output at masked position i.

Magnitude head (regression): A two-layer MLP ($d \to d \to 2$, ReLU activation) outputs $(\mu_i, \log \sigma_i^2)$; the prediction is $\hat{v}_i = \mu_i$.

Sign head (3-class classification): $\hat{s}_i = \mathrm{softmax}(W_{\mathrm{sign}}\, \mathbf{z}_i)$, $W_{\mathrm{sign}} \in \mathbb{R}^{3\times d}$.

Modulo head (100 independent classifiers): For each $m \in \{2, \ldots, 101\}$, a linear layer outputs logits over $\{0, \ldots, m-1\}$ sharing the same input $\mathbf{z}_i$ but with independent parameters. The total output dimension is $\sum_{m=2}^{101} m = 5{,}150$.

3.7 Training Objective

The multi-task loss is:

$$\mathcal{L} = w_{\mathrm{mag}}\, \mathcal{L}_{\mathrm{mag}} + w_{\mathrm{sign}}\, \mathcal{L}_{\mathrm{sign}} + w_{\mathrm{mod}}\, \mathcal{L}_{\mathrm{mod}},$$

with $w_{\mathrm{mag}} = 1.0$, $w_{\mathrm{sign}} = 1.0$, $w_{\mathrm{mod}} = 2.0$ (fixed; adaptive weighting based on uncertainty caused instability in preliminary experiments). $\mathcal{L}_{\mathrm{mag}}$ is Huber loss (Smooth L1) between $\hat{v}_i$ and v_i. $\mathcal{L}_{\mathrm{sign}}$ is cross-entropy over three sign classes. $\mathcal{L}_{\mathrm{mod}}$ is the mean cross-entropy of the 100 Modulo heads, normalised by $\log m$ to correct for varying numbers of classes:

$$\mathcal{L}_{\mathrm{mod}} = \frac{1}{100} \sum_{m=2}^{101} \frac{1}{\log m}\, \mathcal{L}_{\mathrm{CE}}^{(m)}.$$

All losses are computed only at masked positions.

3.8 Baselines

Vanilla Transformer maps each integer to a token ID in a vocabulary of 20,003 entries (values 0–19,999 plus `PAD`, `MASK`, `UNK`); out-of-vocabulary values are replaced by `UNK`. The same three prediction heads are applied to the token embeddings. The vocabulary size 20,003 was chosen so that memory consumption matches IntSeqBERT under the 8 GB VRAM constraint. The prior benchmark FACT [1] handles values up to several million, presupposing larger compute resources than our setting.

Ablation (Magnitude-only) is identical to IntSeqBERT but uses only the Magnitude stream; the FiLM module is removed and $\mathbf{e}_i = \mathbf{h}_i^{\mathrm{mag}}$. This isolates the contribution of the Modulo stream.

3.9 IntegerSolver

The pre-trained model outputs Magnitude $(\mu_i, \log \sigma_i^2)$, sign, and Modulo distributions at masked positions. **IntegerSolver** recovers concrete integers from these predictions.

The Solver derives the 3σ interval $[n_{\min}, n_{\max}]$ from the Magnitude prediction and dynamically selects one of three modes based on the search width $\Delta n = |n_{\max} - n_{\min}|$:

Table 2. Solver operating modes.

Mode	Condition	Method
Dense	$\Delta n \leq 10^6$	Enumerate all integers in range
Sieve	$10^6 < \Delta n \leq 10^{14}$	CRT beam search anchored on high-confidence moduli
CRT	$\Delta n > 10^{14}$	Sparse CRT beam search to generate large integers directly

When the sign prediction is zero, the Solver immediately returns 0. If no valid candidate is found in the search range, the call is recorded as `none`.

Each candidate n is scored as a weighted sum of the Magnitude term and the log-probabilities over all moduli:

$$\text{score}(n) = -\frac{(v_n - \mu_i)^2}{2\sigma_i^2} + 0.3 \cdot \sum_{m=2}^{101} \log P(n \bmod m)\,,$$

where $v_n = 1 + \log_{10}|n|$ $(n \neq 0)$ or 0 $(n = 0)$. The coefficient 0.3 is an empirical hyperparameter that suppresses score inflation due to information overlap between composite and prime moduli. The top k candidates are returned and evaluated as Solver Top-k accuracy (Section 5.4).

4 Dataset and Experimental Setup

4.1 Dataset

As of January 2026, OEIS contains 391,710 sequences. We exclude sequences with fewer than ten terms and sequences carrying any of eleven OEIS keyword tags that are incompatible with integer-sequence prediction, yielding 274,705 sequences (70.1% of the total) as our standard dataset (`std`). Excluded tags fall into three categories: (i) *Out-of-scope representations*: `cons`/`cofr`/`frac` (decimal expansions or fractional values), `base` (base-dependent notation), `word` (linguistic origin); (ii) *Undefined prediction task*: `fini` (finite sequences with no successor term), `tabl` (2-D arrays flattened to 1-D); (iii) *Quality/completeness*: `dead` (deprecated), `unkn`/`less` (incomplete), `dumb` (editorially judged trivial or low-quality). The exclusion of `dumb` restricts the corpus to mathematically meaningful sequences as assessed by OEIS editors.

The corpus is randomly shuffled (seed 42) and split at the sequence level with no overlap in an 8:1:1 ratio:

Each sample corresponds to a sequence prefix of at most $L = 128$ terms (longer sequences are truncated). In the test set, sequence lengths range from 10 to 168 terms (mean 42.5), ensuring that the Solver always receives at least 9 terms of preceding context (Section 5.4).

For scale-stratified evaluation, we define five Magnitude buckets based on $u = \log_{10}|x|$ $(u = 0$ when $x = 0)$:

Table 3. Dataset splits.

Split	Sequences
Training	219,765
Validation	27,470
Test	27,470

Table 4. Magnitude buckets used for scale-stratified evaluation.

Bucket	Range of u	Typical value range	Element fraction
Small	$u < 2$	$\|x\| < 10^2$	52.7%
Medium	$2 \leq u < 5$	$10^2 \leq \|x\| < 10^5$	29.9%
Large	$5 \leq u < 20$	$10^5 \leq \|x\| < 10^{20}$	16.2%
Huge	$20 \leq u < 50$	$10^{20} \leq \|x\| < 10^{50}$	1.1%
Astronomical	$u \geq 50$	$\|x\| \geq 10^{50}$	0.0%

Limitations of the split. The random-shuffle split may place mathematically related sequences (e.g., Fibonacci A000045 and Lucas A000032) in different splits, creating potential leakage. This is a common difficulty of OEIS-based research; a family-aware systematic split is left as future work (Section 7).

4.2 Training Configuration

All models share the following hyperparameters:

Some OEIS values reach $|x| \approx 10^{210}$, causing FP16 overflow; all computation is therefore conducted in FP32. All results are reported by applying the model trained for the fixed 200 epochs to the test set.

4.3 Evaluation Metrics

Magnitude accuracy ($\mathrm{Acc}_{0.5}$): fraction of masked positions satisfying $|\hat{v}_i - v_i| < 0.5$, corresponding to a factor-of-$\sqrt{10} \approx 3.16$ tolerance in the original integer scale.

Sign accuracy: fraction of masked positions where the predicted sign class matches the ground truth.

Mean Modulo Accuracy (MMA): average classification accuracy across all 100 moduli at masked positions.

Solver Top-k accuracy: fraction of next-term prediction instances where the true value appears in the Solver's top-k candidates (Section 5.4).

Normalised Information Gain (NIG): $\mathrm{NIG} = 1 - \mathcal{L}_{\mathrm{CE}}^{(m)} / \ln m$ for modulus m, measuring the information gained about $x \bmod m$ relative to a uniform prior. Higher NIG indicates stronger learned periodic structure for that modulus.

Table 5. Training hyperparameters (common to all models).

Hyperparameter	Value
Epochs	200 (no early stopping)
Batch size	32 (gradient accumulation 2 steps; effective 64)
Learning rate	5×10^{-5}
Warmup fraction	10%
Optimiser	AdamW, weight decay 0.01
Numerical precision	FP32 (AMP disabled)
Mask probability	0.15
GPU	GeForce RTX 3070 Ti (8 GB VRAM) × 1

Table 6. Test results. Mag Acc = $\text{Accuracy}_{0.5}$ (%), Sign Acc (%), MMA = Mean Modulo Accuracy (%). **Bold** indicates the best value within each size group.

Size	Model	Mag Acc	Sign Acc	MMA
Small	**IntSeq**	**94.73**	**97.78**	**40.43**
Small	Vanilla	85.73	96.91	36.21
Small	Ablation	93.72	97.39	25.97
Middle	**IntSeq**	**95.71**	**98.34**	**46.88**
Middle	Vanilla	87.37	97.42	42.53
Middle	Ablation	92.45	97.90	31.93
Large	**IntSeq**	**95.85**	**98.54**	**50.38**
Large	Vanilla	86.97	97.66	45.85
Large	Ablation	89.70	98.29	35.22

5 Experiments

5.1 Main Results

Table 6 shows test-set performance for all model sizes and variants. IntSeqBERT consistently outperforms both baselines on all scales and metrics. At the Large scale, IntSeqBERT surpasses the Vanilla baseline by +8.9 pt in Mag Acc and +4.5 pt in MMA. The Ablation model shows the largest MMA drop at Large scale (−15.2 pt), directly quantifying the contribution of the arithmetic-periodicity features. Notably, the Ablation maintains competitive Mag Acc, indicating that sign and Modulo information contribute only marginally to magnitude regression but are essential for Modulo prediction.

Learning curves. Fig. 2 shows the validation loss across all model sizes and variants. Large IntSeqBERT decreases from Val Loss 2.17 at epoch 1 to 1.01 at epoch 200. Training and validation losses remain closely aligned throughout (epoch 200: Train 1.00 / Val 1.01), indicating no overfitting. Vanilla (final Val Loss 1.77) maintains higher loss than both IntSeqBERT and Ablation (1.39) across the entire training.

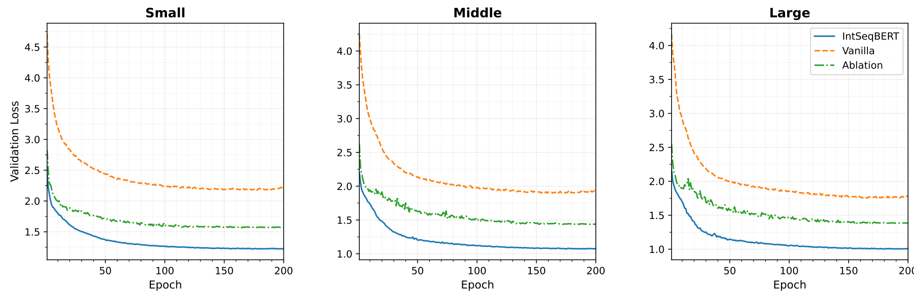

Fig. 2. Validation loss learning curves for all scales (Small / Middle / Large) and all variants. IntSeqBERT (solid blue) consistently achieves lower loss than Vanilla (dashed orange) and Ablation (dash-dot green). At the Large scale, IntSeqBERT converges to Val Loss = 1.01 at epoch 200.

Table 7. Scale-stratified MSE on the test set (Large models). Lower is better.

Bucket	IntSeq	Vanilla	Ablation
Small ($u < 2$)	0.111	0.138	**0.103**
Medium ($2 \leq u < 5$)	**0.051**	0.071	0.116
Large ($5 \leq u < 20$)	**0.162**	2.100	0.381
Huge ($20 \leq u < 50$)	**2.082**	22.73	5.021
Astronomical ($u \geq 50$)	**110.4**	840.0	532.6

5.2 Magnitude Prediction

Scale-stratified analysis. Table 7 reports per-bucket MSE on the test set for Large models. The Vanilla model suffers catastrophic degradation in the Large bucket (MSE = 2.10, 13× that of IntSeqBERT), caused by all out-of-vocabulary integers being absorbed into the `UNK` token. IntSeqBERT achieves the best MSE in all buckets except Small, with particularly pronounced advantage at Medium and above. The Ablation degrades in the Medium bucket without Modulo context (MSE = 0.116 vs. IntSeqBERT 0.051), and collapses more severely in the Huge and Astronomical buckets, indicating that FiLM modulation by the Modulo stream acts as an arithmetic structural constraint on Magnitude estimation for large integers.

Fig. 3 shows predicted vs. true Magnitude scatter plots for the three Large model variants. IntSeqBERT achieves the highest coefficient of determination ($R^2 = 0.988$), substantially reducing off-diagonal dispersion in the Large and Huge buckets compared to Vanilla ($R^2 = 0.943$), visually corroborating the MSE results of Table 7.

Calibration. Fig. 4 shows uncertainty calibration curves for Large models (x-axis: predicted σ; y-axis: actual RMSE; perfect calibration on the diagonal). Vanilla is severely miscalibrated (ECE = 5.36), with overconfidence at low scale ($\sigma \approx 0$ but RMSE $\approx$ 1–2) and exploding σ at high scale. IntSeqBERT (ECE = 0.65) is substantially better and comparable to Ablation (ECE = 0.66), in-

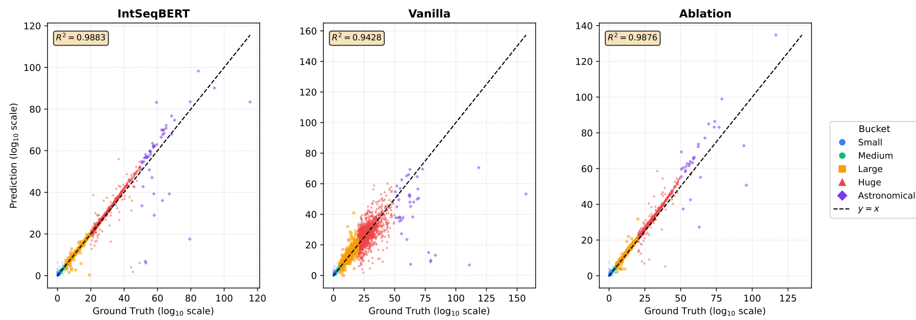

Fig. 3. Predicted vs. true Magnitude ($\log_{10}$ scale) for Large models. Points are coloured by bucket (Small=blue circle, Medium=green circle, Large=yellow-orange square, Huge=red triangle, Astronomical=purple diamond). IntSeqBERT achieves $R^2 = 0.988$ vs. Vanilla $R^2 = 0.943$; Vanilla shows pronounced scatter above the Large bucket.

dicating that calibration improvement is driven primarily by the continuous vs. tokenised representation rather than by the presence of the Modulo stream.

5.3 Modulo Spectrum Analysis

Fig. 5 shows the NIG spectrum for $m = 2, \ldots, 101$ using Large models. IntSeqBERT (solid blue) exceeds both Vanilla and Ablation across the entire spectrum, with a visually clear contrast between prime and composite moduli.

Finding 1: NIG is strongly negatively correlated with Euler's totient ratio. A Pearson correlation of $r = -0.851$ ($p < 10^{-28}$) is observed between NIG and $\varphi(m)/m = \prod_{p|m}(1-1/p)$ (Fig. 6). Composite moduli with many small prime factors achieve higher NIG, consistent with a CRT aggregation effect: when m is a common multiple of smaller moduli $m_1, m_2, \ldots$, then $x \bmod m$ encodes the information of all those moduli simultaneously. The highest NIG across all models and scales is achieved at $m = 96 = 2^5 \times 3$ ($\varphi(96)/96 = 1/3$; Large IntSeq: NIG $= 0.629$, 95% CI [0.622, 0.634]), consistent with $m = 96$ being the largest modulus with totient ratio 1/3 among $\{12, 24, 48, 72, 96\}$ and thus distinguishing the widest value range. The exception is $m = 2$ (prime, NIG $= 0.628$, second highest), reflecting the corpus-wide ubiquity of parity in OEIS sequences.

Finding 2: Parity (mod 2) accuracy stratifies models. Mod-2 accuracy at the Large scale is 85.65% (IntSeq), 81.40% (Vanilla), and 72.13% (Ablation), making parity the single modulus most sensitive to the Modulo stream (-13.5 pt when removed). Table 8 lists representative modulus accuracies.

5.4 Solver-Based Next-Term Prediction

The Solver reconstructs candidate integers from the model's Magnitude, sign, and Modulo predictions and ranks them by likelihood. We evaluate exact-match

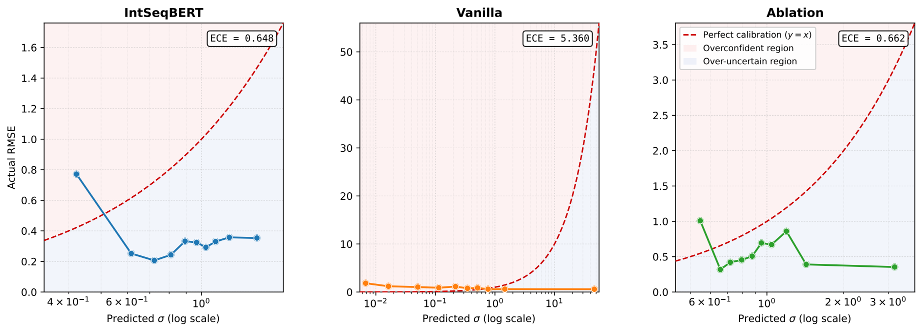

Fig. 4. Uncertainty calibration curves for Large models (x-axis: predicted σ (log scale), y-axis: actual RMSE). The red region (above diagonal) indicates overconfidence; blue indicates overestimation. IntSeqBERT (ECE $= 0.648$) and Ablation (ECE $= 0.662$) lie close to the diagonal, while Vanilla (ECE $= 5.360$) shows severe overconfidence at low σ.

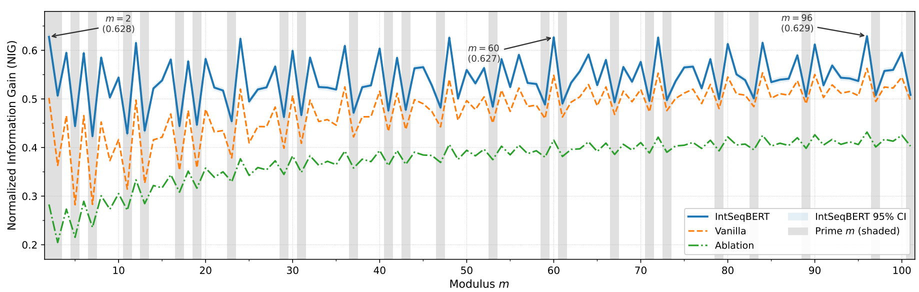

Fig. 5. NIG spectrum for moduli $m = 2, \ldots, 101$ (Large models). Grey shading marks prime moduli. The 95% CI for IntSeqBERT (light blue band) is computed by bootstrapping.

accuracy on 10,000 test samples, where each sample corresponds to one OEIS sequence and the target is the *last term* (all preceding terms are given as context). Sequence lengths in the test set range from 10 to 168 (median 36, mean 42.5), so the Solver always receives at least 9 terms of context.

Valid-candidate rate is the fraction of samples for which the Solver returns at least one candidate. Vanilla always returns a candidate from its vocabulary softmax (rate $= 100\%$). IntSeqBERT and Ablation may fail to find a valid candidate within the search range (`none` mode; 13.4% of Large-IntSeq calls), reducing their valid-candidate rate.

Table 9 shows Solver evaluation results. Large-scale IntSeqBERT achieves a Top-1 accuracy of 19.09%, a **7.4**$\times$ improvement over the Vanilla baseline (2.59%).

Accuracy by Magnitude bucket. Table 10 compares Solver Top-1 accuracy across buckets for all three Large models. IntSeqBERT achieves 68.34%

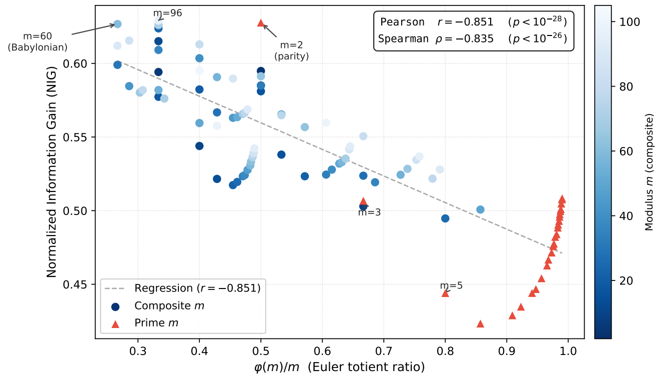

Fig. 6. NIG vs. Euler's totient ratio $\varphi(m)/m$ (Large IntSeqBERT). Composite moduli (blue circles, shade proportional to m) and prime moduli (red triangles) are shown separately. The regression line (grey dashed) indicates Pearson $r = -0.851$ ($p < 10^{-28}$). Notable moduli $m = 2$ (parity), $m = 60$ (Babylonian number), and $m = 96$ (highly composite) are annotated.

Table 8. Representative modulus accuracies for Large models (%).

m	IntSeq	Vanilla	Ablation	Interpretation
2	85.65	81.40	72.13	Parity
3	72.62	65.22	53.72	Ternary residue
5	60.37	50.07	42.63	Last decimal digit parity
10	58.38	49.25	39.47	Least significant decimal digit
60	53.97	47.87	35.12	Babylonian number (highly composite)
96	51.82	47.29	34.44	Highly composite
100	48.51	45.60	33.51	Centesimal residue

Top-1 in the Small bucket, while the Ablation reaches 54.55% and Vanilla only 14.11%. Vanilla collapses to 0% for Medium and above due to the UNK-token degradation of its Magnitude predictions, whereas IntSeqBERT retains meaningful accuracy up to the Medium bucket (20.82%).

Mode breakdown (Large IntSeqBERT). The Solver operates in three modes based on the search width Δn (Section 3.9):

The CRT mode, which attempts to reconstruct large integers directly from Modulo predictions, currently achieves near-zero accuracy (Top-1 = 0.09%). This limitation is discussed in Section 6.3.

Table 9. Solver evaluation: Top-1 and Top-10 exact-match accuracy (%) and valid-candidate rate.

Size	Model	Top-1	Top-10	Sign Acc	Valid rate (%)
Small	**IntSeq**	**14.05**	**21.00**	**98.73**	90.59
Small	Vanilla	2.43	3.24	92.92	100.0
Small	Ablation	7.42	17.33	98.50	90.17
Middle	**IntSeq**	**17.02**	**22.62**	**99.02**	86.31
Middle	Vanilla	2.43	3.41	92.71	100.0
Middle	Ablation	9.88	20.52	98.74	90.34
Large	**IntSeq**	**19.09**	**26.23**	**99.02**	86.64
Large	Vanilla	2.59	3.80	92.05	100.0
Large	Ablation	11.75	21.79	98.94	86.99

Table 10. Solver Top-1 / Top-10 accuracy (%) by Magnitude bucket (Large models). **Bold**: best per bucket.

	IntSeqBERT		Vanilla		Ablation	
Bucket	Top-1	Top-10	Top-1	Top-10	Top-1	Top-10
Small	**68.34**	**88.50**	14.11	20.71	54.55	86.92
Medium	**20.82**	**31.50**	0.00	0.00	5.61	18.88
Large	**0.31**	**0.67**	0.00	0.00	0.03	0.05
Huge	**0.09**	**0.18**	0.00	0.00	0.00	0.00
Astronomical	0.00	0.00	0.00	0.00	0.00	0.00

6 Analysis and Discussion

6.1 Ablation Study: Contribution of the Modulo Stream

To isolate the contribution of the Modulo stream and FiLM fusion, we compare IntSeqBERT with the Magnitude-only Ablation model across all three scales. Table 12 summarises the differences at the Large scale.

As designed, the Modulo stream delivers the largest improvement in Modulo prediction (MMA +15.2 pt, parity +13.5 pt). Its contribution to Magnitude prediction is smaller but consistently significant ($\mathrm{Acc}_{0.5}$ +6.2 pt, MSE -0.228), suggesting that periodic arithmetic structure provides a complementary inductive bias for scale regression: knowing $x_i \bmod m$ for $m \in \{2, \ldots, 101\}$ substantially constrains the set of plausible Magnitude values. The sign accuracy gap is small (+0.25 pt), indicating that sign is largely inferrable from magnitude alone. Solver accuracy improves by +7.3 pt, primarily because more accurate residue information refines scoring in Dense and Sieve modes.

6.2 Scaling Behaviour

Table 13 shows how IntSeqBERT's metrics improve with model size. Magnitude accuracy improves only modestly from Small to Large (+1.1 pt), whereas Modulo accuracy improves substantially (+10.0 pt). This is consistent with the intuition

Table 11. Solver mode breakdown for Large IntSeqBERT.

Mode	Usage (%)	Top-1 (%)	Description
dense	24.0	61.06	Direct integer enumeration
sieve	36.7	5.36	Sieve-based enumeration
crt	23.2	0.09	CRT-based reconstruction
zero	2.7	89.96	Next term predicted as zero
none	13.4	0.00	No valid candidate found

Table 12. IntSeqBERT vs. Ablation at the Large scale (test performance). Δ = IntSeq − Ablation.

Metric	IntSeq	Ablation	Δ
Mag Acc (%)	95.85	89.70	+6.15
Sign accuracy (%)	98.54	98.29	+0.25
MMA (%)	50.38	35.22	+15.16
mod-2 accuracy (%)	85.65	72.13	+13.52
Solver Top-1 (%)	19.09	11.75	+7.34
Magnitude MSE	0.142	0.371	−0.228

that modular arithmetic is more compositional and thus benefits more from increased representational capacity. Solver Top-1 accuracy follows the Modulo trend with consistent improvement (+5.0 pt).

The trends in Table 13 are visualised in Fig. 7, which contrasts the differential scaling of Modulo and Magnitude accuracy across model sizes.

6.3 Limitations

Difficulty with large integers. Solver accuracy collapses to near zero for $|x| \geq 10^{20}$ (Huge and Astronomical buckets). The bottleneck is insufficient Modulo prediction accuracy: CRT-based reconstruction requires *exact* residues for multiple moduli simultaneously, and with MMA at approximately 50%, residue errors are frequent enough to cause reconstruction failure even in the CRT mode (Top-1 = 0.09%).

Low valid-candidate rate. Approximately 13% of IntSeqBERT Solver calls return no valid candidate (`none` mode), primarily due to CRT failures. Recovery may be possible by relaxing residue constraints or adopting approximate CRT in Solver fallback strategies.

Dataset bias. OEIS sequences are predominantly non-negative (tagged `nonn`). Our training and test data inherit this distribution, so the high sign accuracy (98.54%) may partially reflect this bias and could degrade on sequences with many negative terms. Additionally, Astronomical-bucket metrics are based on a small number of samples and should be interpreted with caution.

Table 13. IntSeqBERT scaling: test-split metrics (Small / Middle / Large).

Metric	Small	Middle	Large	Δ (S→L)
Mag Acc (%)	94.73	95.71	95.85	+1.12
MMA (%)	40.43	46.88	50.38	+9.95
mod-2 accuracy (%)	81.97	84.50	85.65	+3.68
Solver Top-1 (%)	14.05	17.02	19.09	+5.04
Magnitude MSE	0.228	0.164	0.142	−0.086

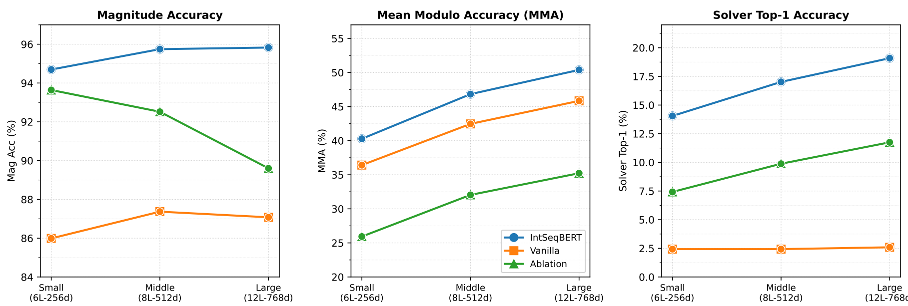

Fig. 7. Scaling behaviour across model sizes (Small / Middle / Large). Left: Magnitude accuracy (Mag Acc) remains high for both IntSeqBERT and Ablation, with only +1.1 pt improvement from Small to Large. Centre: MMA (Mean Modulo Accuracy) for IntSeqBERT improves from 40.3% (Small) to 50.4% (Large, +10.1 pt), consistently exceeding Ablation at all scales. Right: Solver Top-1 accuracy for IntSeqBERT improves from 14.1% to 19.1% (+5.0 pt), with a growing gap over Vanilla (≈2.4%, flat).

7 Conclusion

We have presented **IntSeqBERT**, a dual-stream Transformer encoder for integer sequences. Each sequence element is represented along two complementary axes: a continuous log-scale Magnitude embedding capturing growth behaviour, and a Sin/Cos Modulo embedding capturing periodic arithmetic structure. The two streams are fused via FiLM and jointly trained on 219,765 OEIS sequences with a multi-task objective combining magnitude regression, sign classification, and 100-dimensional Modulo prediction.

On the test set, Large IntSeqBERT achieves 95.85% Magnitude accuracy, 98.54% sign accuracy, and 50.38% MMA, surpassing the Vanilla Transformer by +8.9 pt, +0.9 pt, and +4.5 pt, respectively. For next-term prediction via the Solver, IntSeqBERT achieves Top-1 accuracy of 19.09%, a 7.4× improvement over the Vanilla baseline. Ablation experiments confirm that the Modulo stream accounts for most of the Modulo-prediction gain (+15.2 pt) and also provides a secondary benefit to magnitude regression (+6.2 pt). Cross-modulus analysis reveals a very strong negative correlation between NIG and $\varphi(m)/m$ ($r = -0.851$, $p < 10^{-28}$), providing empirical evidence that composite moduli capture the arithmetic structure of OEIS sequences more efficiently via CRT aggregation.

Future work.

- Combining Magnitude uncertainty estimation with approximate CRT to improve prediction of large integers.
- Extending the Modulo stream beyond $m = 101$ to larger primes and to algebraic structures beyond $\mathbb{Z}/m\mathbb{Z}$.
- Developing finer-grained attention-alignment metrics to identify which recurrence structures each attention head specialises in.
- Constructing family-aware dataset splits that respect mathematical relationships among sequences to prevent leakage between related sequences.
- Extending Solver evaluation to intermediate masked positions, enabling full exploitation of the bidirectional encoder's ability to leverage subsequent context.
- Introducing sequence-synthesis data augmentation as in FACT [1]: adding synthetically generated sequences from recurrence templates and algebraic rules to the training data is expected to improve generalisation, particularly in the sparse Huge and Astronomical buckets.
- Scaling to larger models (hundreds of millions to billions of parameters) and pre-training on the full OEIS corpus, which is expected to yield further gains in Modulo accuracy and CRT-mode performance.

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